PoshanScan: Phase 1 Research Pack

Mid-semester review, October 2026. Every figure below was rechecked against its source on 7 October 2026 unless marked otherwise.

A. Read this first: five things in our proposal that are now out of date or wrong

1. **FSSAI did not withdraw front of pack labelling. It switched to warning labels.** Our proposal and deck say FSSAI "signalled it would withdraw the draft". That was true in January 2026 and is no longer true. On 28 August 2026 FSSAI told the Supreme Court it will use a red hexagon warning for products high in added sugar, added fat or salt, with thresholds from ICMR-NIN 2024. On 10 September it agreed one nutrient alone can trigger the warning. On 28 September 2026 the Court reserved judgment and questioned FSSAI's proposed 365 day voluntary period.
    - Why it matters for us: our argument "India has no label to rely on" becomes "India's label is coming but is at least a year away from being enforced, and a printed label cannot follow you to the next shelf or remember your week". That is a stronger pitch, not a weaker one.
2. **Affordability: "half of India cannot afford a nutritious diet" is outdated.** SOFI 2026 says 35.5% of Indians (520.1 million) could not afford a healthy diet in 2025, down from 59.8% in 2017. SOFI 2025 put it at 42.9% for 2024. Use "more than one in three".
3. **Sweet biscuits are 43% of the chocolate and sugar confectionery category, not 43% of all ultra processed food.** Our deck outline said "Sweet biscuits alone are 43% of it". The WHO and ICRIER report says sweet biscuits are over 43% of retail sales value within chocolate and sugar confectionery. Fix the wording.
4. **NFHS-6 is out (released 30 May 2026). Replace NFHS-5 obesity figures with the newer ones.** Overweight or obese women 15 to 49: 24.0% → 30.7%. Men: 22.9% → 27.3%. Urban women are now at 42.8%.
5. **Chile's warning octagons are black, not red.** Our proposal describes a "red octagon modelled on Chile". Chile uses black. The Supreme Court also flagged in September 2026 that red is read as the non vegetarian mark by Indian shoppers. The prototype uses black for both reasons.

Two figures from our earlier drafts could not be re-verified this week and should be dropped or sourced before they go on a slide: "obesity rose 91% in women and 146% in men between 2005 and 2021", and "ultra processed food is now worth 2.5 trillion rupees".

B. Statistics with sources

Confidence column: **Verified** = checked against the source this week. **Carried** = from our proposal's literature review, source exists but the exact figure was not rechecked this week.

B1. The disease burden

Statistic	Figure	Source	Confidence
Adults with diabetes	101 million (11.4%)	ICMR-INDIAB, Lancet Diabetes & Endocrinology, June 2023	Verified
Adults with prediabetes	136 million (15.3%)	ICMR-INDIAB 2023	Verified
Adults with hypertension	315 million (35.5%)	ICMR-INDIAB 2023	Verified
Generalised obesity	254 million (28.6%)	ICMR-INDIAB 2023	Verified
Abdominal obesity	351 million (39.5%)	ICMR-INDIAB 2023	Verified
High cholesterol / high LDL	213 million / 185 million	ICMR-INDIAB 2023	Verified
Urban vs rural	All metabolic diseases except prediabetes are more common in urban areas	ICMR-INDIAB 2023, survey of 113,043 people across 31 states and UTs, 2008 to 2020	Verified
Women 15 to 49 overweight or obese	20.6% (NFHS-4) → 24.0% (NFHS-5) → 30.7% (NFHS-6)	NFHS-4, 5, 6	Verified
Men 15 to 49 overweight or obese	18.9% → 22.9% → 27.3%	NFHS-4, 5, 6	Verified
Urban women / urban men overweight or obese	42.8% / 36.3%	NFHS-6 (2023 to 24)	Verified
High blood sugar or on medication, men	15.6% → 20.9%	NFHS-5 → NFHS-6	Verified
High blood sugar or on medication, women	13.5% → 17.8%	NFHS-5 → NFHS-6	Verified
Abdominal obesity by waist size	40% of women, 12% of men	Analysis of NFHS-5, Lancet Regional Health Southeast Asia, 2023	Verified

Statistic	Figure	Source	Confidence
Fatty liver (NAFLD), adults	38.6% pooled; 28.2% in community studies, 40.8% in hospital studies	Meta-analysis of 50 Indian studies, J Clin Exp Hepatol 2022	Verified
Fatty liver, children	35.4% pooled; 12.4% in non obese children, 63.4% in obese children	Same meta-analysis	Verified
Share of disease burden from unhealthy diets	56.4%	ICMR-NIN Dietary Guidelines for Indians, May 2024	Verified
Type 2 diabetes preventable by healthy diet and activity	Up to 80%	ICMR-NIN 2024	Verified

Caveat worth saying out loud: the 38.6% fatty liver figure mixes hospital and community samples. The community only figure is 28.2%, which is still more than one in four.

B2. What people eat and buy

Statistic	Figure	Source	Confidence
Ultra processed food retail sales growth	13.37% compound annual growth, 2011 to 2021	WHO and ICRIER report, August 2023	Verified
Sweet biscuits	Over 43% of retail value within chocolate and sugar confectionery (2021)	WHO and ICRIER 2023	Verified
Largest UPF category by volume	Beverages, then chocolate and sugar confectionery, then ready made and convenience food	WHO and ICRIER 2023	Verified
Average salt intake	8.0 g a day (men 8.9 g, women 7.1 g) against a 5 g limit	ICMR-NCDIR National NCD Monitoring Survey, Scientific Reports 2023	Verified
Adults eating under 400 g fruit and vegetables a day	98.4%	National NCD Monitoring Survey	Verified
ICMR-NIN limit on sugar	Under 5% of total energy, which is about 25 g (6 teaspoons) on a 2,000 kcal diet	ICMR-NIN 2024	Verified

B3. Can people afford the swap?

Statistic	Figure	Source	Confidence
People unable to afford a healthy diet, 2025	520.1 million, 35.5% of the population	SOFI 2026 (FAO, IFAD, UNICEF, WFP, WHO)	Verified
Same, 2017	59.8%	SOFI 2026	Verified
Cost of a healthy diet in India	PPP $4.11 per person per day in 2025, up about 48% since 2017	SOFI 2026	Verified

B4. Do people read labels?

Statistic	Figure	Source	Confidence
Supermarket shoppers in Delhi and Hyderabad who read labels	90%	Vemula et al., ICMR-NIN, Public Health Nutrition	Verified
Of those, who only check the manufacturing or expiry date	81%	Same	Verified
South Delhi urban village: read labels / of those, only check dates	64.1% / 86%	National Medical Journal of India, 2018	Verified

B5. What works and what does not

Statistic	Figure	Source	Confidence
Nudges that change the act of choosing vs nudges that give information	Behavioural nudges cut about 209 kcal a day, cognitive (information) nudges about 64 kcal: roughly three times more effective	Cadario and Chandon, Marketing Science 2020; 299 effect sizes from 96 field experiments	Verified
Reducing unhealthy eating vs increasing healthy eating	Nudges work better at cutting unhealthy eating	Cadario and Chandon 2020	Verified
Chile's labelling law	Purchases of "high in" beverages fell 23.7%, larger than standalone sugar taxes in Latin America	Taillie et al., PLOS Medicine 2020; 2,383 urban households	Verified
Chile thresholds (solids per 100 g)	Added sugar 10 g, sodium 400 mg, saturated fat 4 g, energy 275 kcal	Law 20.606, final stage, June 2019	Verified
Chile thresholds (liquids per 100 ml)	Sugar 5 g, sodium 100 mg, saturated fat 3 g, energy 70 kcal	Same	Verified
Default order vs offering swaps (online supermarket RCT, 1,088 UK shoppers)	Reordering cut saturated fat in the basket by 5.0%, swaps alone by 2.0%	Koutoukidis et al., IJBNPA 2019	Verified
Health app dropout	43% pooled (40% in RCTs, 49% in observational studies)	Meyerowitz-Katz et al., JMIR 2020	Verified
mDiabetes, India	56 text messages in 12 languages over 6 months; better fruit, vegetable and fat intake, no change in exercise	Pfammatter et al., JMIR 2016	Verified
Digital nutrition interventions in LMICs	Improve knowledge and some behaviours; body measurements rarely change	Kurniawan et al., J Health Popul Nutr 2025 (scoping review)	Carried
Nudge effects shrink after publication bias correction	Effect largely disappears for some domains	Maier et al., PNAS 2022, reanalysing Mertens et al. 2022	Carried

B6. The channel

Statistic	Figure	Source	Confidence
WhatsApp monthly users in India	Over 500 million (Meta's own figure, December 2024). Third party 2026 estimates range from 535 to 854 million	Meta via TechCrunch; DataReportal	Verified (use "over 500 million")

B7. India's labelling timeline

Date	Event
Sep 2022	FSSAI drafts the Indian Nutrition Rating, a 0.5 to 5 star system
9 Apr 2025	Supreme Court directs the government to finalise front of pack labelling within three months
30 Jan 2026	FSSAI affidavit: no stakeholder consensus on star ratings, more research needed
10 Feb 2026	Supreme Court urges FSSAI to consider warning labels instead
13 Aug 2026	Supreme Court: "If you can't do it, we will"
28 Aug 2026	FSSAI proposes a red hexagon warning for products high in two or more of added sugar, added fat, salt, using ICMR-NIN 2024 thresholds, in phases
10 Sep 2026	FSSAI agrees one nutrient can trigger the warning. Court questions red (non veg association), font size, and per 100 g vs per serve
28 Sep 2026	Court reserves judgment. FSSAI wants 365 days of voluntary compliance after notification

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C. Gap analysis

1. **Nobody stands at the moment of purchase in India.** The strong choice architecture experiments are Dutch, British, Australian and Chilean. We found no published Indian trial testing a point of purchase nudge on packaged food. Our review in the proposal is the basis for this; it should be phrased as "we found no" rather than "none exists".

2. **The label gap is closing, but slowly.** India's warning label is likely at least a year from enforcement after notification, and a printed label cannot compare products, suggest a swap or notice a pattern. A scan tool aligned to the same ICMR-NIN thresholds can work now and stay consistent once the label arrives.
3. **People "read" labels without reading the nutrition panel.** Nine in ten say they read labels; eight in ten of those only look at dates. Any tool that hands people a nutrition table repeats the failure. A single symbol plus a physical number (teaspoons) is the evidence based fix.
4. **Information alone does not move behaviour, and apps get deleted.** Behavioural nudges outperform information about three to one, and health apps lose 43% of users. That is why we use a channel people already open daily and why we intervene before the purchase, not after the meal.
5. **Swaps must not cost more.** With more than one in three Indians unable to afford a healthy diet, a recommendation that raises the bill is a lecture, not a nudge. Every swap in the prototype is priced at or below the scanned item where possible.
6. **The shopper is not always the decision maker.** In many households the person holding the packet did not choose the list or control the budget. This is a known limit of an individual nudge; a household group version is a Phase 3 research question, not a claim.

D. Status against our timeline

Phase	Planned end	Status
1. Secondary research, statistics with citations, gap analysis	First week of September	Done. This pack, with five corrections flagged in section A
2. Prototype of the message flow: scan, warning, swap	First week of October	Done. Working chat prototype with rules engine, 7 day scan memory and pattern nudge; full build spec ready for the software version
3. Software and final report	Second week of November	Next: real label data for 20 products, Hindi messages, WhatsApp sandbox, align thresholds with FSSAI's final notification, ethics section

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E. Source links

- ICMR-INDIAB 2023 summary: https://nmji.in/lessons-learnt-from-the-icmrindiab-study
- ICMR-INDIAB figures (IANS): https://morungexpress.com/315-mn-indians-suffer-from-hypertension-101-mn-are-diabetic-icmr-study

- NFHS-6 key findings: https://vajiramandravi.com/current-affairs/national-family-health-survey-6/
- NFHS-5 obesity: https://www.downtoearth.org.in/news/health/obesity-increases-in-india-haryana-karnataka-among-worst-performers-nfhs-5-80381
- NFHS-5 abdominal obesity: https://pubmed.ncbi.nlm.nih.gov/37492420/
- NAFLD meta-analysis: https://pmc.ncbi.nlm.nih.gov/articles/PMC9168741
- ICMR-NIN Dietary Guidelines 2024 coverage: https://thesouthfirst.com/news/icmr-releases-dietary-guidelines-for-indians-after-13-years-says-no-to-protein-supplements
- WHO and ICRIER ultra processed food report: https://www.icrier.org/pdf/PPT_WHO-report-release_Aug17_2023.pdf and https://www.downtoearth.org.in/news/health/india-s-ultra-food-processing-sector-growing-who-calls-for-nutrient-based-tax-model-91336
- Salt intake, National NCD Monitoring Survey: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10517942/
- SOFI 2026 India findings: https://currentaffairs.khanglobalstudies.com/the-state-of-food-security-and-nutrition-in-the-world-sofi-2026/
- Vemula et al., label use: https://pmc.ncbi.nlm.nih.gov/articles/PMC11108719/
- Label use, South Delhi: https://nmji.in/use-of-information-on-pre-packaged-foods-among-residents-of-an-urban-village-of-south-delhi-india/
- Cadario and Chandon 2020: https://knowledge.insead.edu/marketing/which-healthy-eating-nudges-work-best
- Chile beverage purchases, Taillie et al. 2020: https://pmc.ncbi.nlm.nih.gov/articles/PMC7012389
- Chile thresholds, Law 20.606: https://www.carey.cl/en/entry-into-force-of-the-last-stage-of-law-no-20606
- Koutoukidis et al. 2019: https://doaj.org/article/8e36fa80a68a44b9848830b50aacebf9
- App dropout, Meyerowitz-Katz et al. 2020: https://www.jmir.org/2020/9/e20283/
- mDiabetes: https://pmc.ncbi.nlm.nih.gov/articles/PMC4992169
- Kurniawan et al. 2025: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12492931/
- WhatsApp India users: https://richautomate.in/blog/whatsapp-users-india-2026-statistics
- FSSAI and Supreme Court: https://www.livelaw.in/top-stories/why-one-year-needed-to-place-front-of-pack-labels-on-packaged-food-supreme-court-asks-fssai-reserves-judgment-552122 , https://www.livelaw.in/top-stories/will-consider-warning-labels-on-foods-high-in-any-single-nutrient-of-concern-fssai-revises-stand-in-supreme-court-549577 , https://indialegallive.com/constitutional-law-news/courts-news/supreme-court-seeks-clarity-from-centre-fssai-on-phased-front-of-pack-warning-labels/ , https://foxmandal.in/News/sc-urges-fssai-to-consider-front-of-pack-warning-labels/